A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Conservation priorities, legislation & protocols - Set 1

Mark scheme

AQA spec 3.1.2.3.1-3.1.2.3.2; The living environment; textbook pp. 26-39

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 3 points below. **1** Coordinating global data on biodiversity conservation (Red List) **1** Increasing understanding of the importance of biodiversity **1** Deploying nature-based solutions to global challenges in climate, food and sustainable development **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	3

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 3 points below. **1** Decline = (12 000 - 2100) / 12 000 x 100 = 82.5% **1** Decline over 80% meets the Critically Endangered criterion even though habitat area (400 km^2) and adults (2100) only meet Endangered **1** A species is placed in the highest category for which it meets any one criterion, so Critically Endangered **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
01.3	**1 mark each** 1 mark for each of the 3 points below. **1** Categorising a species requires a great deal of research (population, range, trends) and resources – about 85 000 of millions of species assessed **1** Key species in habitats under particular threat (eg lemurs indicating the state of Madagascar's forests) **1** Evolutionarily unique species / EDGE species – endemic species – keystone species – flagship species – species with fragmented populations **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **endemic** → found naturally only in that one place • **keystone species** → a species whose loss changes the whole community	3
01.4	1 mark for the point below. **1 Iberian lynx:** Critically Endangered to Endangered after captive breeding and protection of two habitat areas (population trebled in 15 years) – Fregate Island beetle after rat eradication **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **captive breeding** → breeding in captivity to increase numbers for release **Do not credit** • captive breeding alone as conservation - credit needs reintroduction, habitat protection or genetic management as well	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 26-27; AQA spec 3.1.2.3.1.	

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Question	Marking guidance	Marks
02.1	**2 marks per point** 2 marks for each of the 4 points below; award 1 mark where a point is only partly made. **2 EDGE:** Evolutionarily Distinct and Globally Endangered - few close relatives, often the last species of their genus / family (Bactrian camel, pygmy hippo, northern bald ibis) – loss would remove a unique branch of evolution / genes **2 Endemic:** found in only one area (often islands - Galapagos, Madagascar, Seychelles: Aldabra giant tortoise, red ruffed lemur) – if lost locally there is no population elsewhere to recolonise **2 Keystone:** disproportionately important for the structure of the community relative to its abundance - predation of dominant species, seed dispersal, habitat creation (jaguar, forest elephants keeping paths and water holes open, wolves, beavers) – losing them causes loss of many other species **2 Flagship:** high public profile (tiger, panda, elephant, orang-utan) raising funds and support that also protects the whole habitat and less charismatic species (fungi, bats, moths) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Endemic** → found naturally only in that one place	8
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** A fragmented population may consist of isolated groups each too small to be viable (resources, inbreeding) even if total numbers are large **1** A species confined to a single area (eg bonobos in one country) is vulnerable to any local change / conflict, unlike chimpanzees spread across many countries **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 28-29; AQA spec 3.1.2.3.1.	
03.1	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Designation of protected areas (SSSIs, MCZs) **1** Protection of wild birds and their nests (except some pests / game birds) **1** Protection of named mammals such as otters, dormice, red squirrels, badgers and their setts **1** Uprooting wild plants generally illegal **1** Bats must not be disturbed – roof timber treatments must be non-toxic to bats **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks.	4

Do not write outside the box

Question	Marking guidance	Marks
03.2	**1 mark each** 1 mark for each of the 4 points below. **1 Appendix I:** species threatened with extinction - all international (commercial) trade banned, except movement for conservation breeding **1 Examples:** great apes, big cats, rhinos, blue whale **1 Appendix II:** species that may become threatened if trade is not closely controlled - trade permitted under licence from countries where the species is well protected **1 Examples:** Honduras mahogany, hippopotamus, green iguana, great white / basking shark, Venus fly trap **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **CITES** → the treaty that controls international trade in endangered species **Do not credit** • 'CITES bans all trade in wildlife' - it controls and licenses trade, with different rules for each appendix	4
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** Collection may be hard to police in remote areas / poor countries, but if the animals or products cannot be exported / imported there is no market **1** Removing the profit removes the incentive to collect **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 30-31; AQA spec 3.1.2.3.2.	

Do not write outside the box

Question	Marking guidance	Marks
04.1	**1 mark each** 1 mark for each of the 4 points below. **1 SSSI:** UK legislation (Wildlife and Countryside Act) – best examples of UK flora, fauna, geological features – management plan agreed with owner who must notify Operations Likely to Damage (ploughing, drainage, pesticides, tree planting / removal) **1 SPA:** EU Birds Directive – places of international importance for birds (with SACs under the Habitats Directive form the Natura 2000 network; may restrict activities in surrounding areas) **1 Ramsar:** Ramsar Convention (intergovernmental agreement) – internationally important wetlands, originally for waterfowl, now wetland conservation and wise use generally **1 MCZ:** UK marine legislation (Marine and Coastal Access Act) under the W&CA framework – protects marine habitats / species by restricting activities such as fishing and dredging – NNR: best SSSIs with complete communities **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	4
04.2	**1 mark each** 1 mark for each of the 3 points below. **1 Legal status:** restrictions on damaging activities within the area (and sometimes in surrounding areas, eg drainage) **1** Management agreements between owners and statutory bodies to maintain suitable conditions **1** But it does not protect against external influences such as air pollution or climate change – very few sites are state-owned so protection depends on owner cooperation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
04.3	**1 mark each** 1 mark for each of the 3 points below. **1** Much of the UK's wildlife lives in habitats created by farming (hay meadows, hedgerows) which only survive if the farming continues **1** Managing land for wildlife costs money / reduces income so farmers need payments (points for beetle banks, hedgerows, buffer strips, wild bird seed, low-input grassland) **1** Benefits everyone so society pays through central funding – legislation cannot compel positive management **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 30, 38-39; AQA spec 3.1.2.3.2.	

Do not write outside the box

Question	Marking guidance	Marks
05.1	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** (240 000 - 2 000) / 240 000 x 100 = 99.2% **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1
05.2	**1 mark each** 1 mark for each of the 3 points below. **1 Whales are k-selected:** slow growing, late maturing, one calf every few years, so populations recover very slowly **1** Technology (harpoon cannons, steam / factory ships) allowed catches far above the sustainable yield within decades **1** No regulation / open access to the ocean – whales were valuable (oil, spermaceti) and easily found **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	3
05.3	**1 mark each** 1 mark for each of the 4 points below. **1** Moratorium on commercial whaling since 1986 until populations recover **1** Total protection for certain species – whale sanctuaries **1** Limits on numbers and sizes taken – protection of suckling mothers and calves **1** Research into whale biology and threats (ship strikes, net entanglement) – exceptions: aboriginal subsistence quotas (eg Inuit bowheads), 'scientific' permits (Japan - ruled not research in 2014), Norway / Iceland set their own quotas **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4

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Question	Marking guidance	Marks
05.4	**1 mark each** 1 mark for each of the 2 points below. **1** ITTO's 1990 target of trade only from sustainably managed forests by 2000 was not met and little progress since **1** Timber comes from many countries / private land with weak enforcement, high demand and alternative land uses – whaling involves few nations and a single treaty body **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 31-32; AQA spec 3.1.2.3.2.	
06.1	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** The size of the population, and the number of mature individuals able to breed **1** The rate of change of the population: how fast numbers have fallen, usually over the last ten years or three generations **1** The geographical range the species occupies and how far it is fragmented or dispersed, since a small or divided range makes extinction more likely **1** The quality and area of the remaining habitat, and whether the threats acting on the species are continuing **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
06.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** (12 000 - 2 100) / 12 000 x 100 **1** = 82.5% (accept 82 - 83%) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • Points after the first 2 - the question asks for 2, so a list of extras cannot pick up the marks. • A bare wrong answer with no working.	2

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Question	Marking guidance	Marks
06.3	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** It is a keystone species, so losing it would change the whole community and many other species would be lost with it **1** It is endemic to the area, so if it is lost there it is lost everywhere – a species found in many countries can be conserved elsewhere **1** It is a flagship or charismatic species that attracts public interest and funding, which also protects everything else living in its habitat **1** It is a genetic resource with known or likely value: a wild relative of a crop, or a source of a medicine **1** Conservation is practical and likely to succeed: the threats can be removed, and suitable habitat remains for a reintroduction **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **keystone species** → a species whose loss changes the whole community • **endemic** → found naturally only in that one place **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
06.4	**1 mark each** 1 mark for each of the 3 points below. **1** Appendix I lists species threatened with extinction, for which commercial international trade is banned and only exceptional permits such as scientific exchange are issued **1** Appendix II lists species that are not yet threatened with extinction but may become so unless trade is controlled, so trade is allowed under licence and monitored **1** Where the threat is demand in distant wealthy markets, cutting off the market removes the incentive to poach, and the agreement acts in every signatory country rather than only where the habitat is protected **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **CITES** → the treaty that controls international trade in endangered species **Do not credit** • 'CITES bans all trade in wildlife' - it controls and licenses trade, with different rules for each appendix	3

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Question	Marking guidance	Marks
06.5	**Any 4 of 6** 1 mark each for any 4 of the 6 points below, in any order, maximum 4. **1** Sites of Special Scientific Interest are notified for their wildlife or geology, and damaging operations on them need consent from the conservation agency **1** National Nature Reserves and Local Nature Reserves are owned or managed for conservation, with management plans and wardens **1** The Wildlife and Countryside Act makes it an offence to kill, injure or take listed species, or to damage their breeding sites and resting places **1** Special Areas of Conservation and Special Protection Areas, and Ramsar sites for wetlands, give international protection to habitats and to migratory birds **1** Green belts, National Parks and Areas of Outstanding Natural Beauty restrict development and so protect habitat indirectly **1** Agri-environmental schemes pay farmers to manage hedgerows, field margins and meadows for wildlife on land that is still farmed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks.	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 26-32, 38-39; AQA spec 3.1.2.3.1-3.1.2.3.2.	

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